

Supermarket Sales Analysis

Objective:

The objectives for data analysis in your supermarket sales project should focus on understanding patterns, trends, and opportunities to support data-driven decision-making.

Problem Statement:

Supermarket often struggle to optimize product placement, forecast demand, and identify high-performing products. This project aims to analyze the supermarket sales data. Identify the most and least popular products. Understand seasonal trends and peak sales periods. Determine the relationship between sales and customer demographics. Suggest ways to optimize inventory and marketing strategies.

What is the total sales and total profit? What is the average sales & profit per order? Which product category generates the most sales and which generates the most profit? Which product sub-category has the highest average sales & profit? Which customer segment (Consumer, Corporate, Home Office) generates the most sales & profit? Which state generates the most sales & Profit? Which city generates the most sales & profit? What is the total sales & profit by each ship mode? What is the total sales & profit by each product? What is the total sales & profit by each region? What are the total sales and profit over time year? How does the sales trend look over the months/years? Which months have the highest sales in 2014, 2015, 2016, 2017? What is the total sales and profit by customer (Customer ID or Customer Name)? Which customers have made the highest number of orders? How many products in each sub-category are sold in total (Quantity)?

Install & Import Necessary Libraries

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Loading Dataset

```
In [7]: # download from https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
# Load the CSV file into a pandas DataFrame
# Path to your Excel file
file_path = 'D:/Anudip/Python/Project/Book2.xlsx'
df = pd.read_excel(file_path)
```

```
In [9]: # Display the first 5 rows of the DataFrame
df.head(6)
```

Out[9]:

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City	State	Postal Code	Region	Product ID	Category	Category Name
0	CA-2016-152156	2016-08-11	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Books
1	CA-2016-152156	2016-08-11	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Clothing
2	CA-2016-161389	2016-05-12	2016-10-12	Standard Class	IM-15070	Irene Maddox	Consumer	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Birds
3	CA-2016-117590	2016-08-12	2016-10-12	First Class	GH-14485	Gene Hale	Corporate	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones
4	CA-2016-117590	2016-08-12	2016-10-12	First Class	GH-14485	Gene Hale	Corporate	Richardson	Texas	75080	Central	FUR-FU-10003664	Furniture	Furniture
5	CA-2014-106376	2014-05-12	2014-10-12	Standard Class	BS-11590	Brendan Sweed	Corporate	Gilbert	Arizona	85234	West	OFF-AR-10002671	Office Supplies	

Understand the Structure of the Data

```
In [8]: # Checking overview of data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2739 entries, 0 to 2738
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order ID              2739 non-null   object
1   Order Date            2739 non-null   datetime64[ns]
2   Ship Date            2739 non-null   datetime64[ns]
3   Ship Mode             2739 non-null   object
4   Customer ID           2739 non-null   object
5   Customer Name         2739 non-null   object
6   Segment              2739 non-null   object
7   City                 2739 non-null   object
8   State                2739 non-null   object
9   Postal Code          2739 non-null   int64
10  Region               2739 non-null   object
11  Product ID           2739 non-null   object
12  Category             2739 non-null   object
13  Sub-Category         2739 non-null   object
14  Product Name         2739 non-null   object
15  Sales                2739 non-null   float64
16  Quantity             2739 non-null   int64
17  Discount             2739 non-null   float64
18  Profit               2739 non-null   float64
19  Total Sales          2739 non-null   float64
dtypes: datetime64[ns](2), float64(4), int64(2), object(12)
memory usage: 428.1+ KB
```

```
In [9]: # Used to generate descriptive statistics for the numerical columns in a DataFrame
# Quick summary of key statistical measures such as count, mean, standard deviation, minimum, and maximum values, as well as q
df.describe()
```

Out[9]:

	Order Date	Ship Date	Postal Code	Sales	Quantity	Discount	Profit	Total Sales
count	2739	2739	2739.000000	2739.000000	2739.000000	2739.000000	2739.000000	2739.000000
mean	2016-02-05 17:53:33.581598976	2016-05-22 15:33:42.782037248	54220.505294	230.971407	3.767798	0.158759	31.325717	1122.454815
min	2014-01-02 00:00:00	2014-02-05 00:00:00	1040.000000	0.556000	1.000000	0.000000	-3839.990400	-0.244000
25%	2015-03-11 00:00:00	2015-07-12 00:00:00	22178.500000	17.902000	2.000000	0.000000	1.828050	51.897000
50%	2016-04-07 00:00:00	2016-07-12 00:00:00	53142.000000	58.240000	3.000000	0.200000	8.773500	189.800000
75%	2017-02-12 00:00:00	2017-06-11 00:00:00	90004.000000	200.387500	5.000000	0.200000	30.599900	730.875000
max	2017-12-11 00:00:00	2017-12-12 00:00:00	99207.000000	17499.950000	14.000000	0.800000	8399.976000	87499.750000
std	NaN	NaN	32183.801502	629.384413	2.205670	0.206339	244.168288	3662.541871

```
In [10]: # Checking rows & columns present in data frame
df.shape
```

Out[10]: (2739, 20)

```
In [11]: df.columns
```

```
Out[11]: Index(['Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID',
               'Customer Name', 'Segment', 'City', 'State', 'Postal Code', 'Region',
               'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales',
               'Quantity', 'Discount', 'Profit', 'Total Sales'],
              dtype='object')
```

Data Cleaning

```
In [19]: # Found DataFrame with datetime columns, Convert columns from datetime to date
```

```
In [13]: # Convert columns from datetime to date (removes the time part)
df['Order Date'] = df['Order Date'].dt.date
```

```
In [15]: df['Ship Date'] = df['Ship Date'].dt.date
```

```
In [19]: # Check the DataFrame and column data types  
print(df)  
print(df.dtypes)
```

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID \
0	CA-2016-152156	2016-08-11	2016-11-11	Second Class	CG-12520
1	CA-2016-152156	2016-08-11	2016-11-11	Second Class	CG-12520
2	CA-2016-161389	2016-05-12	2016-10-12	Standard Class	IM-15070
3	CA-2016-117590	2016-08-12	2016-10-12	First Class	GH-14485
4	CA-2016-117590	2016-08-12	2016-10-12	First Class	GH-14485
...	...	...	...	...	...
2734	US-2016-103674	2016-06-12	2016-10-12	Standard Class	AP-10720
2735	US-2016-103674	2016-06-12	2016-10-12	Standard Class	AP-10720
2736	US-2015-151435	2015-06-09	2015-09-09	Second Class	SW-20455
2737	CA-2017-163566	2017-03-08	2017-06-08	First Class	TB-21055
2738	CA-2017-119914	2017-04-05	2017-09-05	Second Class	CC-12220

	Customer Name	Segment	City	State	Postal Code \
0	Claire Gute	Consumer	Henderson	Kentucky	42420
1	Claire Gute	Consumer	Henderson	Kentucky	42420
2	Irene Maddox	Consumer	Seattle	Washington	98103
3	Gene Hale	Corporate	Richardson	Texas	75080
4	Gene Hale	Corporate	Richardson	Texas	75080
...	...	...	...	...	...
2734	Anne Pryor	Home Office	Los Angeles	California	90032
2735	Anne Pryor	Home Office	Los Angeles	California	90032
2736	Shaun Weien	Consumer	Lafayette	Louisiana	70506
2737	Ted Butterfield	Consumer	Fairfield	Ohio	45014
2738	Chris Cortes	Consumer	Westminster	California	92683

	Region	Product ID	Category	Sub-Category \
0	South	FUR-BO-10001798	Furniture	Bookcases
1	South	FUR-CH-10000454	Furniture	Chairs
2	West	OFF-BI-10003656	Office Supplies	Binders
3	Central	TEC-PH-10004977	Technology	Phones
4	Central	FUR-FU-10003664	Furniture	Furnishings
...	...	...	...	...
2734	West	OFF-BI-10003727	Office Supplies	Binders
2735	West	OFF-BI-10002026	Office Supplies	Binders
2736	South	FUR-TA-10001039	Furniture	Tables
2737	East	OFF-LA-10004484	Office Supplies	Labels
2738	West	OFF-AP-10002684	Office Supplies	Appliances

	Product Name	Sales	Quantity \
0	Bush Somerset Collection Bookcase	261.960	2

1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.940	3
2	Fellowes PB200 Plastic Comb Binding Machine	407.976	3
3	GE 30524EE4	1097.544	7
4	Electrix Architect's Clamp-On Swing Arm Lamp, ...	190.920	5
...	...	...	...
2734	Avery Durable Slant Ring Binders With Label Ho...	13.376	4
2735	Ibico Recycled Linen-Style Covers	437.472	14
2736	KI Adjustable-Height Table	85.980	1
2737	Avery 476	16.520	5
2738	Acco 7-Outlet Masterpiece Power Center, Wihtou...	243.160	2

	Discount	Profit	Total Sales
0	0.0	41.9136	523.920
1	0.0	219.5820	2195.820
2	0.2	132.5922	1223.728
3	0.2	123.4737	7682.608
4	0.6	-147.9630	954.000
...	...	...	...
2734	0.2	4.6816	53.304
2735	0.2	153.1152	6124.408
2736	0.0	22.3548	85.980
2737	0.2	5.3690	82.400
2738	0.0	72.9480	486.320

[2739 rows x 20 columns]

Order ID	object
Order Date	object
Ship Date	object
Ship Mode	object
Customer ID	object
Customer Name	object
Segment	object
City	object
State	object
Postal Code	int64
Region	object
Product ID	object
Category	object
Sub-Category	object
Product Name	object
Sales	float64


```
Quantity      int64
Discount      float64
Profit        float64
Total Sales   float64
dtype: object
```

```
In [25]: # Important part of data analysis, check duplicates
# Using Conditional statement
if df.duplicated().sum() > 0:
    print('Duplicates are present in dataframe')
else:
    print('Duplicate does not exist')
```

Duplicate does not exist

```
In [27]: # Another method of check duplicates
df.duplicated().sum()
```

Out[27]: 0

```
In [29]: # Check for null values
df.isnull().sum()

# Remove rows with null values in the column
# df.dropna(subset=['Column Name'])
# Remove duplicates based on the column
# df.drop_duplicates(subset=['Column Name'])
```

```
Out[29]: Order ID      0
        Order Date    0
        Ship Date     0
        Ship Mode     0
        Customer ID   0
        Customer Name  0
        Segment       0
        City          0
        State         0
        Postal Code   0
        Region        0
        Product ID    0
        Category      0
        Sub-Category  0
        Product Name   0
        Sales         0
        Quantity      0
        Discount      0
        Profit        0
        Total Sales   0
        dtype: int64
```

Exploratory Data Analysis (EDA)

```
In [32]: # What is the total sales and total profit?
        df['Total Sales'].sum().round()
```

```
Out[32]: 3074404.0
```

```
In [34]: df['Profit'].sum().round()
```

```
Out[34]: 85801.0
```

```
In [36]: # What is the average sales per order?
        # Count the number of unique orders
        UniqueOrders = df['Order ID'].nunique()
        print(UniqueOrders)
```

1358

```
In [38]: # Calculate the average sales per order
# AverageSalesPerOrder = Total Sales / UniqueOrders or using groupby
AverageSalesPerOrder = df.groupby('Order ID')['Total Sales'].sum().mean().round()
print(f"Average Sales per Order: ${AverageSalesPerOrder:,.2f}")
```

Average Sales per Order: \$2,264.00

```
In [223]: # Calculate the average Profit per order
# AverageProfitPerOrder = Total Sales / UniqueOrders or using groupby
AverageProfitPerOrder = df.groupby('Order ID')['Profit'].sum().mean().round()
print(f"Average Profit per Order: ${AverageProfitPerOrder:,.2f}")
```

Average Profit per Order: \$63.00

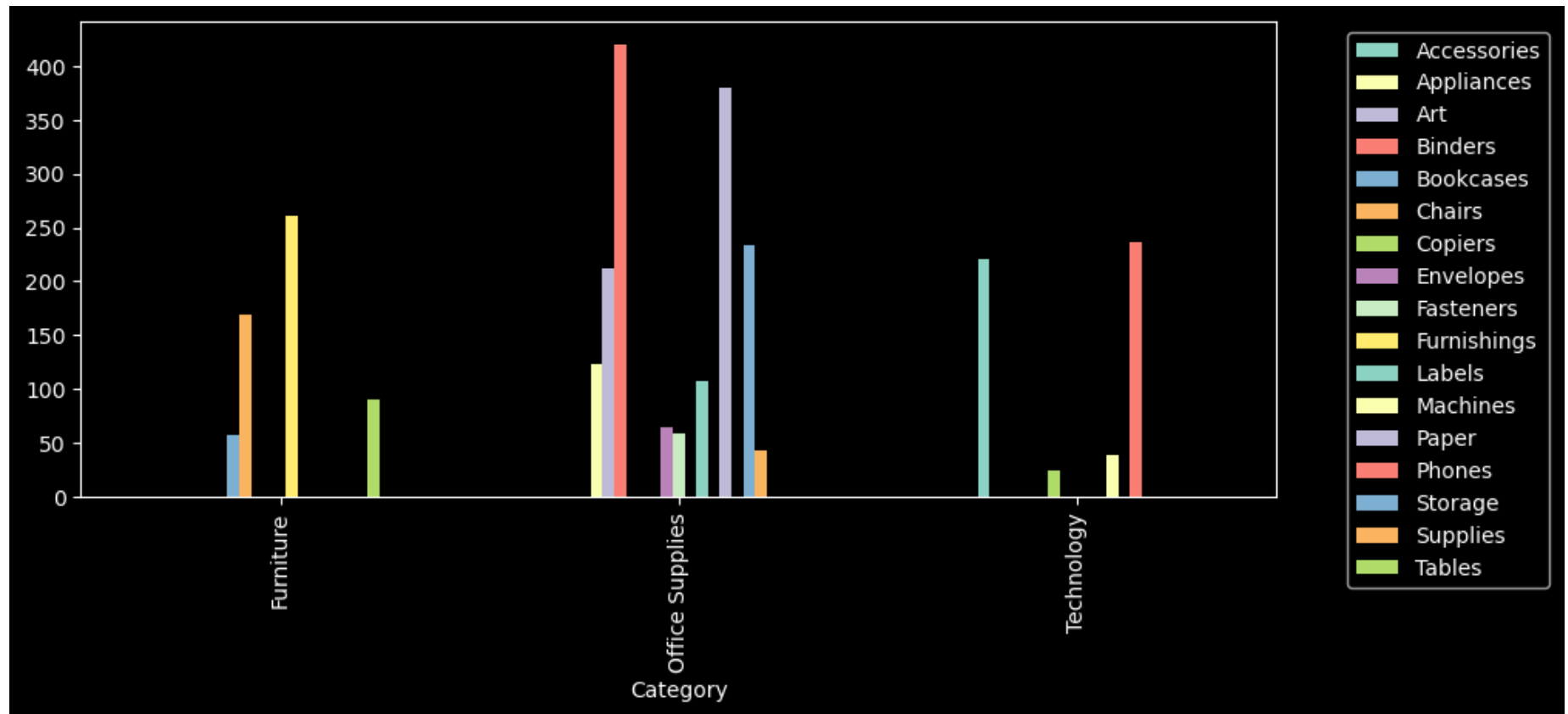
```
In [40]: df['Category'].value_counts()
```

```
Out[40]: Category
Office Supplies    1642
Furniture           577
Technology          520
Name: count, dtype: int64
```

```
In [42]: df['Sub-Category'].value_counts()
```

```
Out[42]: Sub-Category
Binders      421
Paper        380
Furnishings  261
Phones       237
Storage      234
Accessories  221
Art          212
Chairs       169
Appliances   123
Labels       107
Tables        90
Envelopes    64
Fasteners    58
Bookcases    57
Supplies     43
Machines     38
Copiers      24
Name: count, dtype: int64
```

```
In [166... # sub-categories have higher counts of products
pd.crosstab(df['Category'], df['Sub-Category']).plot(kind='bar', stacked=False, figsize=(10, 4))
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```



```
In [45]: # Which product category generates the most sales and which generates the most profit?
CategorySalesProfit = df.groupby('Category')[['Total Sales', 'Profit']].sum().round()
print(CategorySalesProfit)
```

	Total Sales	Profit
Category		
Furniture	1040408.0	8026.0
Office Supplies	819439.0	27022.0
Technology	1214556.0	50754.0

```
In [164]: # Which product category generates the most sales and which generates the most profit?
# Create a figure with subplots
fig, axes = plt.subplots(1, 2, figsize=(10, 4)) # 1 row, 2 columns

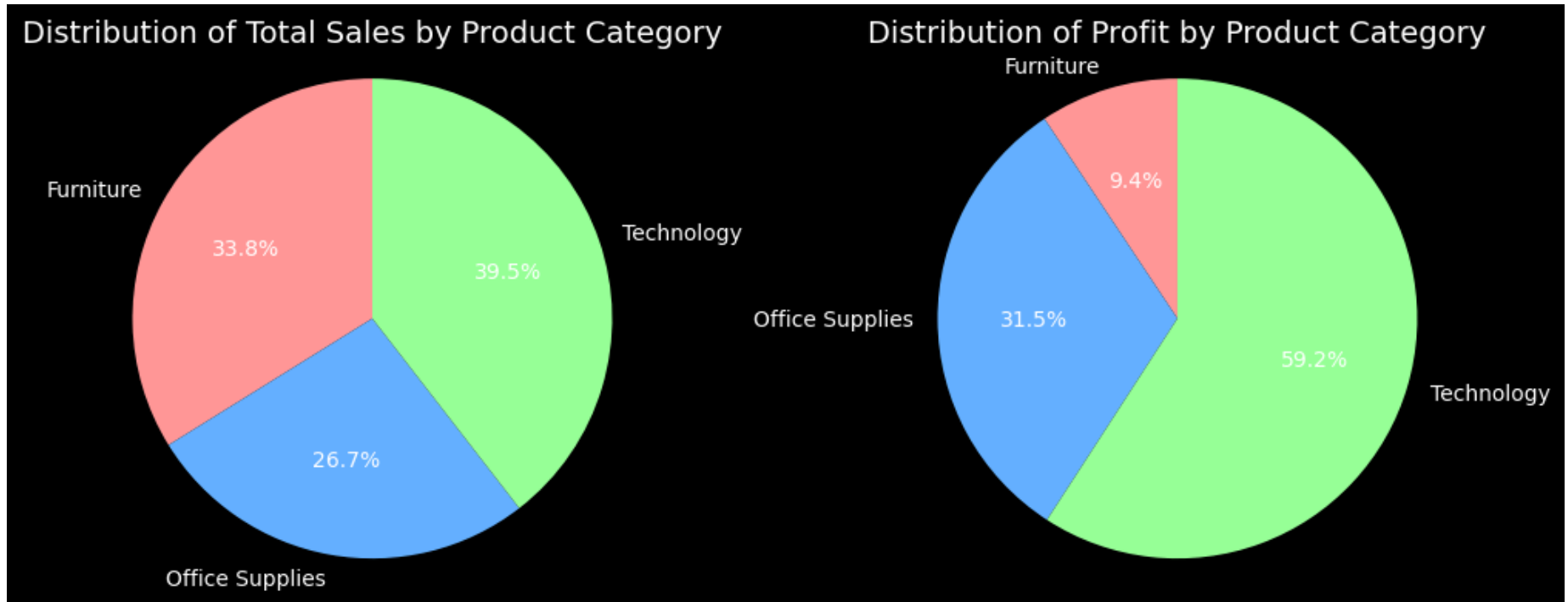
# Pie chart for Total Sales
```

```
axes[0].pie(CategorySalesProfit['Total Sales'],
            labels=CategorySalesProfit.index,
            autopct='%1.1f%%',
            startangle=90,
            colors=['#ff9999','#66b3ff','#99ff99'])
axes[0].set_title('Distribution of Total Sales by Product Category', fontsize=14)
axes[0].axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

# Pie chart for Profit
axes[1].pie(CategorySalesProfit['Profit'],
            labels=CategorySalesProfit.index,
            autopct='%1.1f%%',
            startangle=90,
            colors=['#ff9999','#66b3ff','#99ff99'])
axes[1].set_title('Distribution of Profit by Product Category', fontsize=14)
axes[1].axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

# Adjust layout for better spacing
plt.tight_layout()

# Show the plots
plt.show()
```



```
In [50]: # Which product sub-category has the highest average sales?
AverageSales_Subcategory = df.groupby('Sub-Category')[['Total Sales', 'Profit']].mean().round()

# Sort by Total Sales in descending order to find the highest average sales
AverageSales_Subcategory_sorted = AverageSales_Subcategory.sort_values(by='Total Sales', ascending=False)

# Print the sorted result
print(AverageSales_Subcategory_sorted)
```

Sub-Category	Total Sales	Profit
Copiers	9686.0	809.0
Machines	9073.0	154.0
Tables	3094.0	-38.0
Chairs	2861.0	59.0
Bookcases	2791.0	-15.0
Phones	1613.0	52.0
Accessories	1154.0	60.0
Appliances	1107.0	19.0
Storage	1083.0	28.0
Supplies	1043.0	-23.0
Binders	475.0	11.0
Furnishings	457.0	9.0
Envelopes	265.0	26.0
Paper	263.0	24.0
Art	212.0	9.0
Labels	176.0	15.0
Fasteners	79.0	3.0

```
In [158... # Create a figure with subplots (1 row, 2 columns)
fig, axes = plt.subplots(1, 2, figsize=(10, 4)) # 1 row, 2 columns

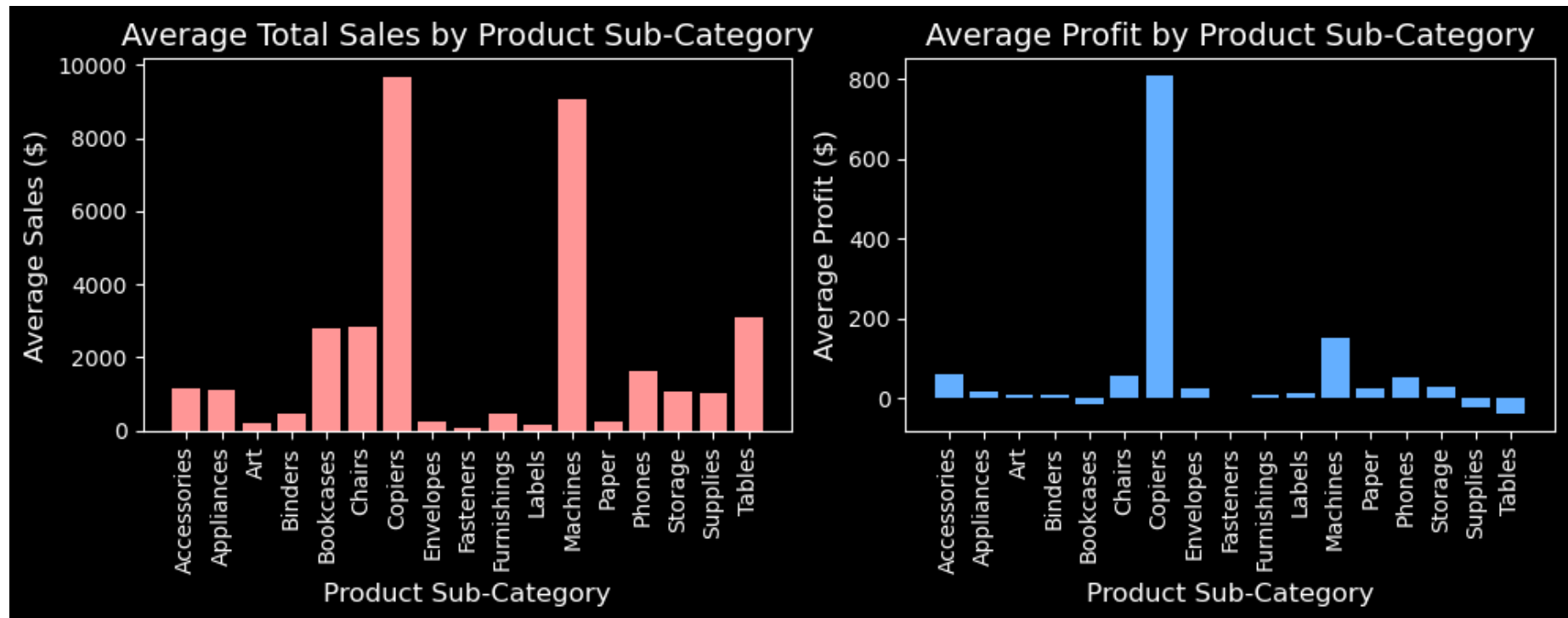
# Bar chart for Average Total Sales
axes[0].bar(AverageSales_Subcategory.index, AverageSales_Subcategory['Total Sales'], color='#ff9999')
axes[0].set_title('Average Total Sales by Product Sub-Category', fontsize=14)
axes[0].set_xlabel('Product Sub-Category', fontsize=12)
axes[0].set_ylabel('Average Sales ($)', fontsize=12)
axes[0].tick_params(axis='x', rotation=90) # Rotate x-axis labels for better readability

# Bar chart for Average Profit
axes[1].bar(AverageSales_Subcategory.index, AverageSales_Subcategory['Profit'], color='#66b3ff')
axes[1].set_title('Average Profit by Product Sub-Category', fontsize=14)
axes[1].set_xlabel('Product Sub-Category', fontsize=12)
axes[1].set_ylabel('Average Profit ($)', fontsize=12)
axes[1].tick_params(axis='x', rotation=90) # Rotate x-axis labels for better readability

# Adjust layout for better spacing
plt.tight_layout()
```



```
# Show the plots
plt.show()
```



```
In [168... # Which customer segment (Consumer, Corporate, Home Office) generates the most sales?
SegmentWise = df.groupby('Segment')[['Total Sales', 'Profit']].sum().round()
print(SegmentWise)
```

Segment	Total Sales	Profit
Consumer	1358477.0	34271.0
Corporate	994148.0	26574.0
Home Office	721779.0	24956.0

```
In [170... # Reset index to convert 'Segment' into a column and drop the unnecessary 'level_0' column if it exists
SegmentWise = SegmentWise.reset_index()

# Check the columns of the DataFrame
print(SegmentWise.columns) # Ensure 'Segment' exists in the DataFrame
```

```
# Create a new column 'Category' to represent the hierarchical structure
SegmentWise['Category'] = 'Sales' # This will be a constant value for each row

# Create the sunburst plot
fig = px.sunburst(
    SegmentWise,
    path=['Category', 'Segment'], # Hierarchy: first "Category", then "Segment"
    values='Total Sales', # The size of the segment is based on Total Sales
    hover_data=['Profit'], # Display Profit when hovering
    title="Sales by Customer Segment"
)
# Update the layout for better display
fig.update_layout(
    width=600, # Set the width of the plot
    height=500, # Set the height of the plot
    title={'x': 1.0} # Optional: center the title
)
fig.show()
```

```
Index(['Segment', 'Total Sales', 'Profit'], dtype='object')
```

```
In [231... # Which state generates the most sales?  
StateWise = df.groupby('State')[['Total Sales', 'Profit']].sum().round()  
StateWise.sort_values('Total Sales', ascending=False)  
print(StateWise.head(10))
```

	Total Sales	Profit
State		
Alabama	11215.0	394.0
Arizona	31100.0	317.0
Arkansas	15598.0	1668.0
California	636730.0	22395.0
Colorado	34708.0	-1192.0
Connecticut	16973.0	1150.0
Delaware	19341.0	1523.0
District of Columbia	271.0	13.0
Florida	83512.0	-684.0
Georgia	37617.0	2300.0

```
In [172... # Group by 'State' and calculate the sum of 'Total Sales' and 'Profit'
StateWise = df.groupby('State')[['Total Sales', 'Profit']].sum().round()

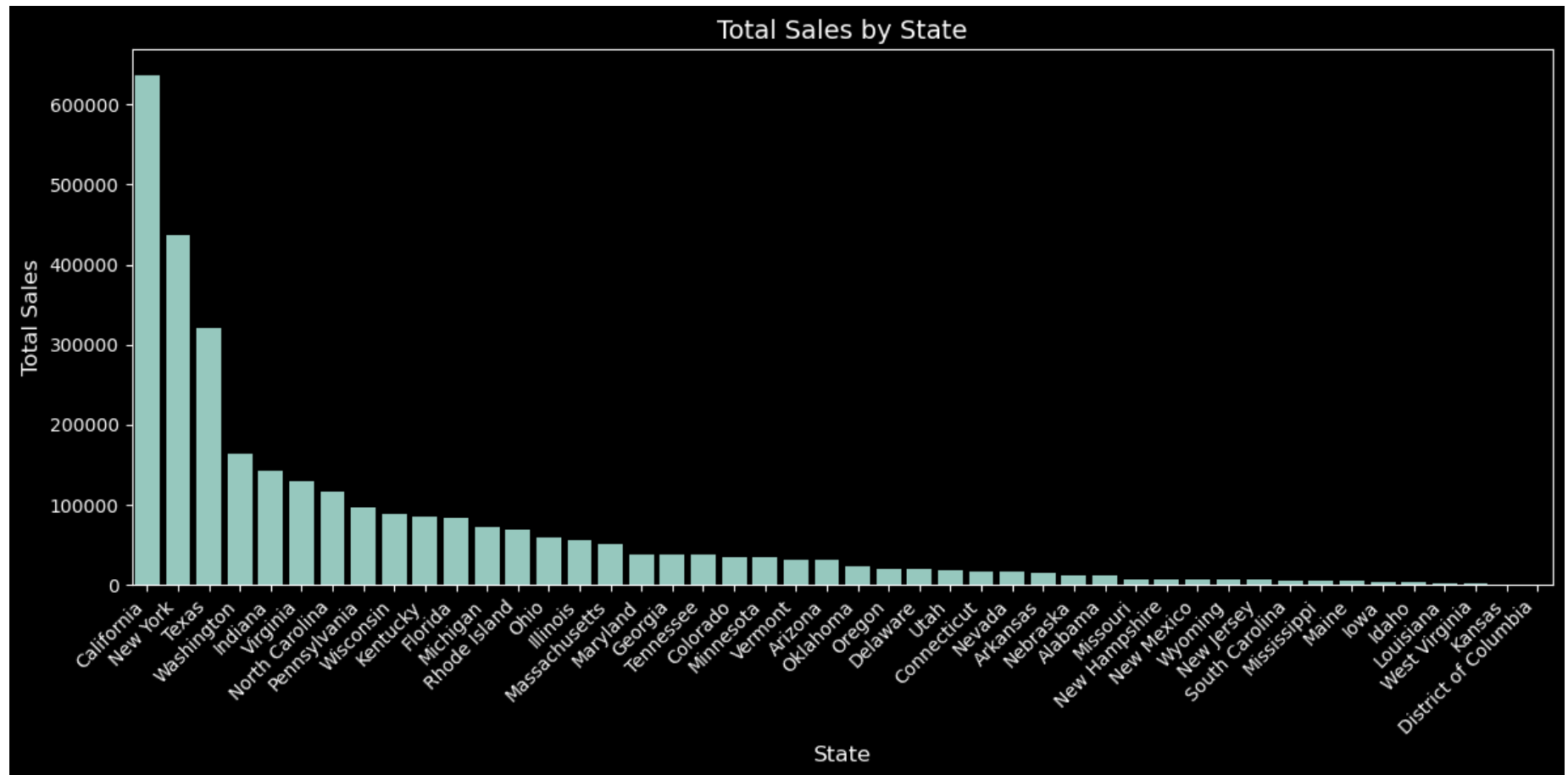
# Sort the 'StateWise' DataFrame by 'Total Sales' in descending order
StateWise_sorted = StateWise.sort_values('Total Sales', ascending=False)

# Create a bar plot for total sales by state
plt.figure(figsize=(12, 6))
sns.barplot(x=StateWise_sorted.index, y='Total Sales', data=StateWise_sorted)

# Adding labels and title
plt.title('Total Sales by State', fontsize=14)
plt.xlabel('State', fontsize=12)
plt.ylabel('Total Sales', fontsize=12)

# Rotate x-axis labels for better visibility
plt.xticks(rotation=45, ha='right')

# Display the plot
plt.tight_layout()
plt.show()
```



```
In [249... # Which city generates the most sales?
CityWise = df.groupby('City')[['Total Sales', 'Profit']].sum().round()
CityWise.sort_values('Total Sales', ascending=False)
print(CityWise.head(10))
```

	Total Sales	Profit
City		
Akron	214.0	-19.0
Albuquerque	1791.0	113.0
Alexandria	23937.0	188.0
Allen	551.0	-26.0
Amarillo	1970.0	-75.0
Anaheim	11210.0	272.0
Andover	2047.0	124.0
Ann Arbor	4445.0	229.0
Antioch	58.0	9.0
Apple Valley	181.0	24.0

```
In [176... # Group by 'City' and calculate the sum of 'Total Sales' and 'Profit'
CityWise = df.groupby('City')[['Total Sales', 'Profit']].sum().round()

# Sort the 'CityWise' DataFrame by 'Total Sales' in descending order
CityWise_sorted = CityWise.sort_values('Total Sales', ascending=False)

# Select top cities for better visualization (optional, if too many cities)
CityWise_top = CityWise_sorted.head(10)

# Create an area plot
plt.figure(figsize=(12, 6))

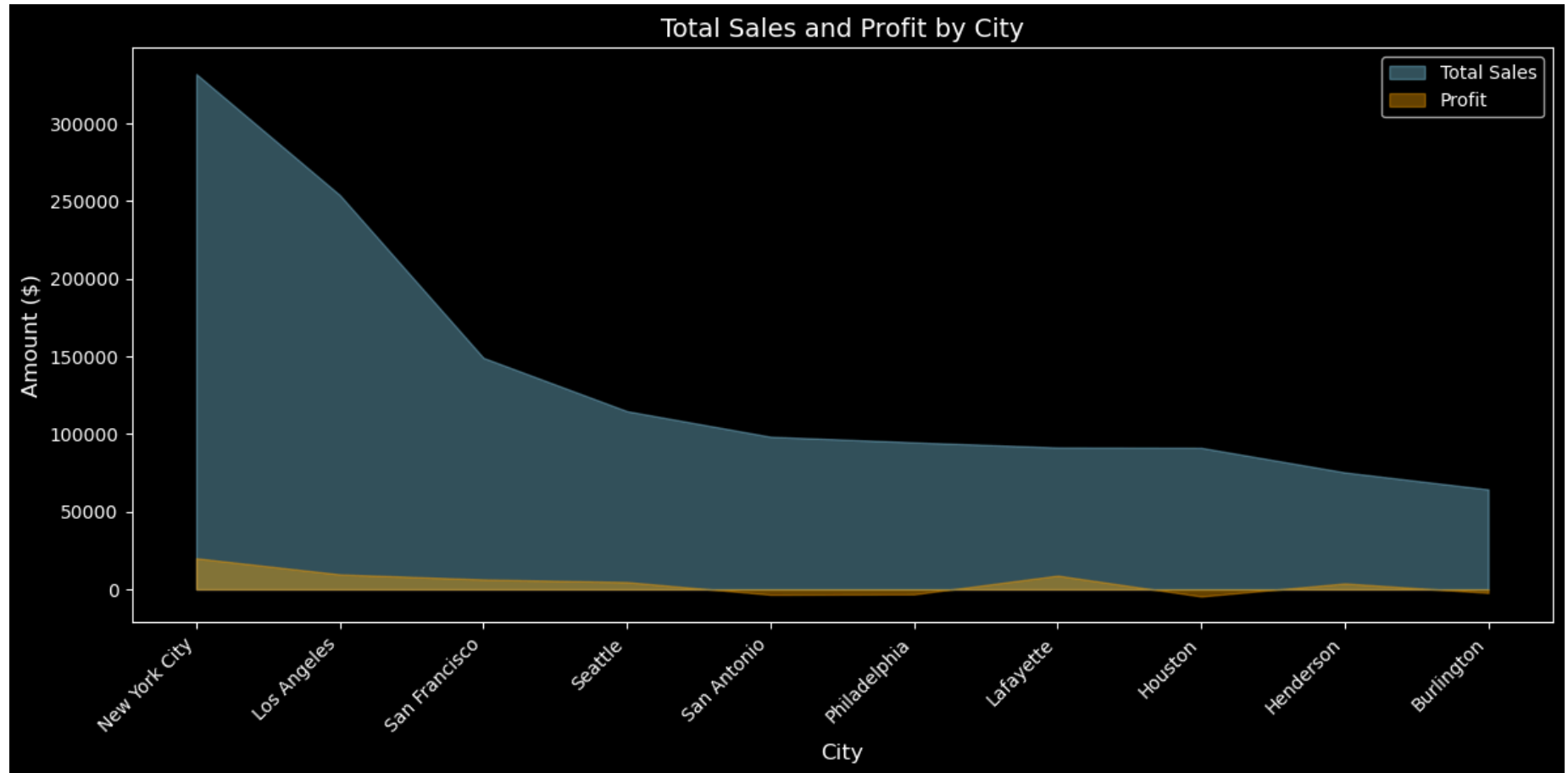
# Plot area plot for Total Sales
plt.fill_between(CityWise_top.index, CityWise_top['Total Sales'], color="skyblue", alpha=0.4, label="Total Sales")
# Plot area plot for Profit
plt.fill_between(CityWise_top.index, CityWise_top['Profit'], color="orange", alpha=0.4, label="Profit")

# Adding labels and title
plt.title('Total Sales and Profit by City', fontsize=14)
plt.xlabel('City', fontsize=12)
plt.ylabel('Amount ($)', fontsize=12)

# Rotate x-axis labels for better visibility
plt.xticks(rotation=45, ha='right')

# Display the legend
plt.legend()
```

```
# Display the plot  
plt.tight_layout()  
plt.show()
```



```
In [62]: # What is the total sales & profit by each ship mode?  
ShipModeWise = df.groupby('Ship Mode')[['Total Sales', 'Profit']].sum().round()  
print(ShipModeWise)
```

	Total Sales	Profit
Ship Mode		
First Class	538333.0	14234.0
Same Day	262970.0	6172.0
Second Class	653032.0	19647.0
Standard Class	1620069.0	45748.0

In [156...

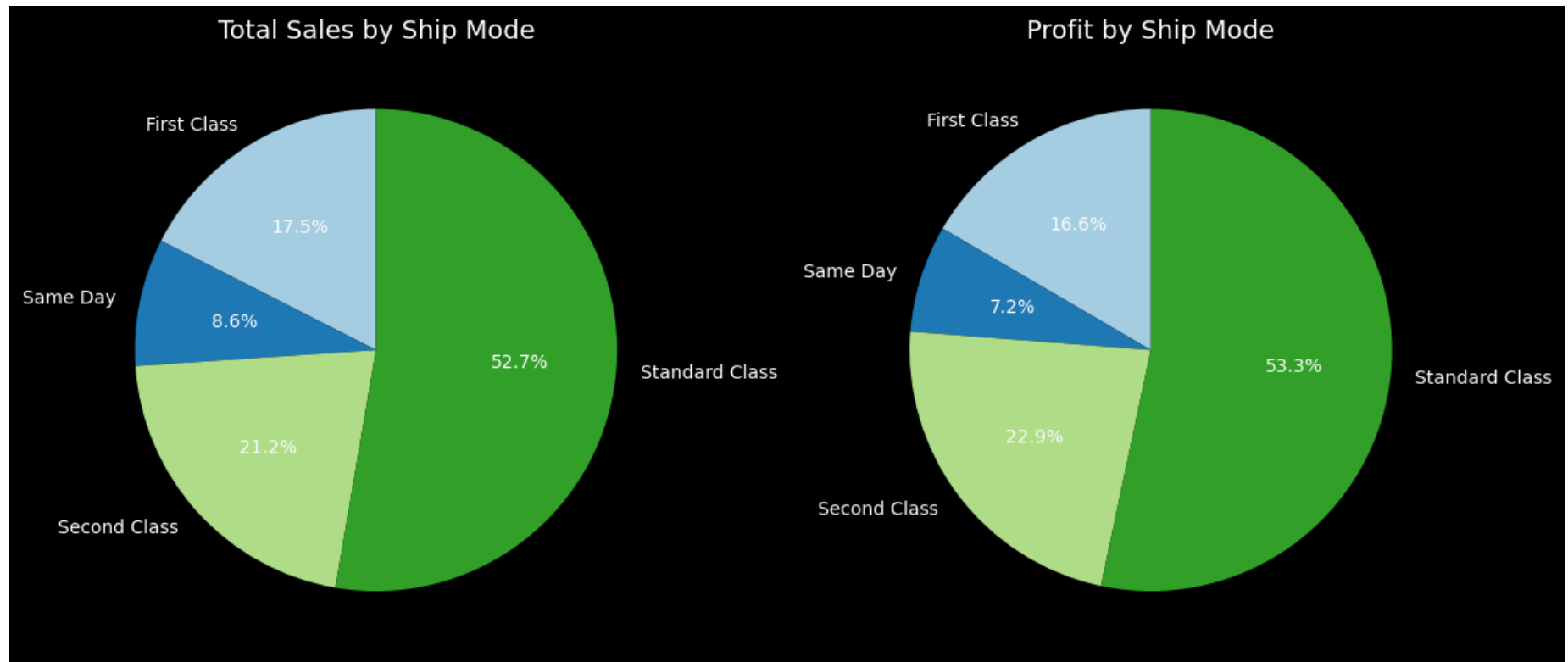
```
# Group by 'Ship Mode' and calculate the sum of 'Total Sales' and 'Profit'
ShipModeWise = df.groupby('Ship Mode')[['Total Sales', 'Profit']].sum().round()

# Create subplots (1 row, 2 columns)
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

# Pie chart for Total Sales by Ship Mode
axes[0].pie(ShipModeWise['Total Sales'], labels=ShipModeWise.index, autopct='%1.1f%%', startangle=90, colors=plt.cm.Paired.colors)
axes[0].set_title('Total Sales by Ship Mode', fontsize=14)

# Pie chart for Profit by Ship Mode
axes[1].pie(ShipModeWise['Profit'], labels=ShipModeWise.index, autopct='%1.1f%%', startangle=90, colors=plt.cm.Paired.colors)
axes[1].set_title('Profit by Ship Mode', fontsize=14)

# Display the pie charts
plt.tight_layout() # Ensures the pie charts don't overlap
plt.show()
```

```
In [257... # What is the total sales & profit by each product?
ProductWiseSales = df.groupby('Product Name')['Total Sales'].sum().round().sort_values(ascending=False).head(5)
print(ProductWiseSales)
```

```
Product Name
Canon imageCLASS 2200 Advanced Copier      87500.0
Lexmark MX611dhe Monochrome Laser Printer  71399.0
Hewlett Packard LaserJet 3310 Copier        59518.0
Honeywell Enviracaire Portable HEPA Air Cleaner for 17' x 22' Room  44196.0
HP Designjet T520 Inkjet Large Format Printer - 24" Color  43750.0
Name: Total Sales, dtype: float64
```

```
In [253... ProductWiseProfit = df.groupby('Product Name')['Profit'].sum().round().sort_values(ascending=False).head(5)
print(ProductWiseProfit)
```

```

Product Name
Canon imageCLASS 2200 Advanced Copier                8400.0
Hewlett Packard LaserJet 3310 Copier                  3168.0
HP Designjet T520 Inkjet Large Format Printer - 24" Color  2800.0
Ativa V4110MDD Micro-Cut Shredder                    2401.0
Honeywell Enviracaire Portable HEPA Air Cleaner for 17' x 22' Room  2282.0
Name: Profit, dtype: float64

```

```

In [68]: # What is the total sales & profit by each region?
RegionWise = df.groupby('Region')[['Total Sales', 'Profit']].sum().round()
print(RegionWise)

```

Region	Total Sales	Profit
Central	762800.0	13350.0
East	840971.0	32964.0
South	531325.0	10400.0
West	939308.0	29087.0

```

In [154... # Group by 'Region' and calculate the sum of 'Total Sales' and 'Profit'
RegionWise = df.groupby('Region')[['Total Sales', 'Profit']].sum().round()

# Data for plotting
regions = RegionWise.index
total_sales = RegionWise['Total Sales']
profit = RegionWise['Profit']

# Create a figure
plt.figure(figsize=(10, 6))

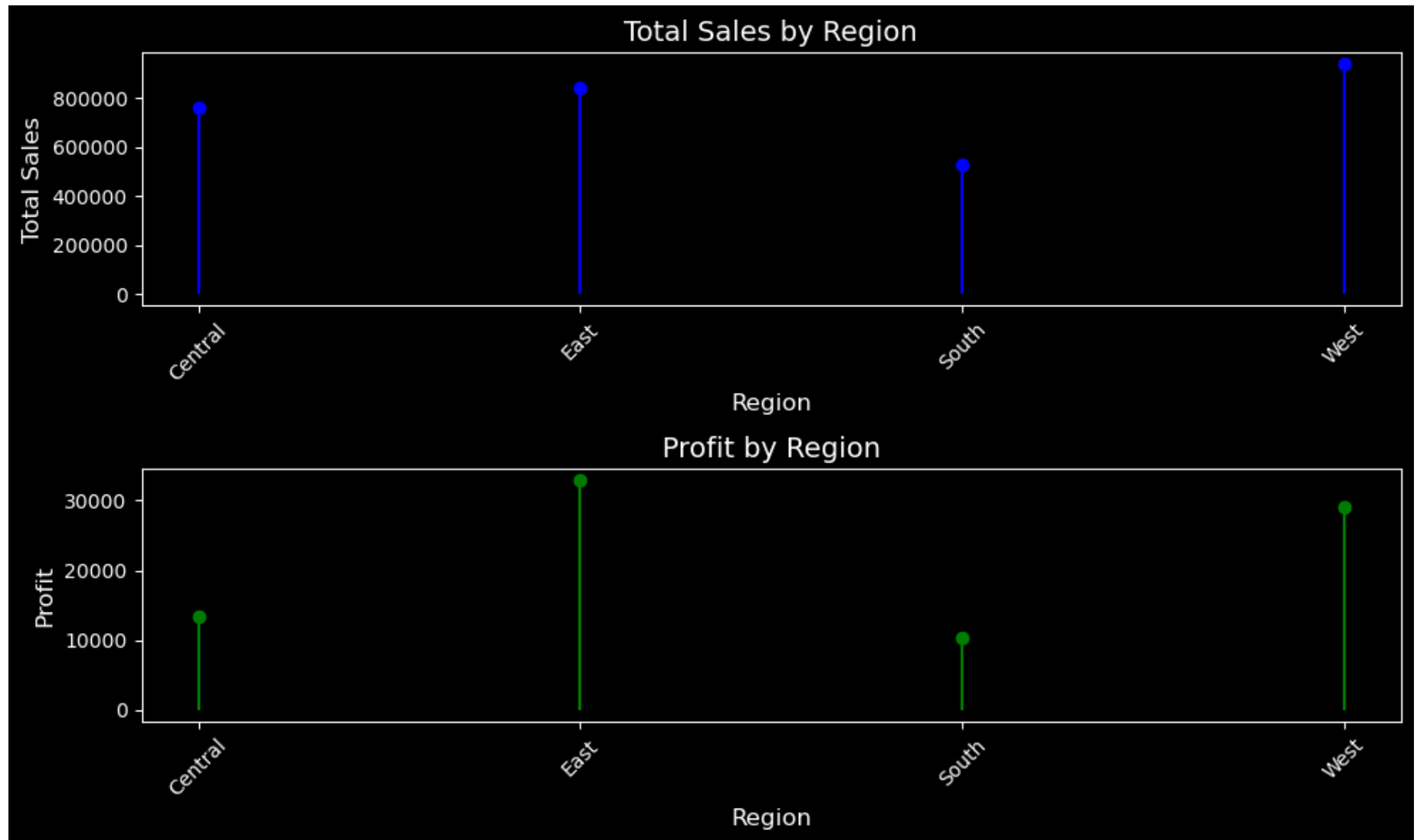
# Create a stem plot for Total Sales
plt.subplot(2, 1, 1) # 2 rows, 1 column, first plot
plt.stem(regions, total_sales, basefmt=" ", linefmt='b-', markerfmt='bo', label='Total Sales')
plt.title('Total Sales by Region', fontsize=14)
plt.xlabel('Region', fontsize=12)
plt.ylabel('Total Sales', fontsize=12)
plt.xticks(rotation=45)

# Create a stem plot for Profit
plt.subplot(2, 1, 2) # 2 rows, 1 column, second plot
plt.stem(regions, profit, basefmt=" ", linefmt='g-', markerfmt='go', label='Profit')

```

```
plt.title('Profit by Region', fontsize=14)
plt.xlabel('Region', fontsize=12)
plt.ylabel('Profit', fontsize=12)
plt.xticks(rotation=45)

# Display the plots
plt.tight_layout()
plt.show()
```



Time-based Analysis

```
In [73]: # What are the total sales and profit over time year?
# Convert 'Order Date' to datetime format
df['Order Date'] = pd.to_datetime(df['Order Date'])
```

```
# Extract Year and Month from 'Order Date'
df['Year'] = df['Order Date'].dt.year

# Group by Year and Month and calculate total sales and profit
SalesProfit_by_Year = df.groupby('Year')[['Total Sales', 'Profit']].sum().round()

# Print the result
print(SalesProfit_by_Year)
```

	Total Sales	Profit
Year		
2014	544583.0	13820.0
2015	583515.0	12556.0
2016	980794.0	36505.0
2017	965512.0	22921.0

```
In [265... # How does the sales trend look over the months/years?
# Convert 'Order Date' to datetime format if it's not already
df['Order Date'] = pd.to_datetime(df['Order Date'])

# Extract the 'Year' and 'Month' from 'Order Date'
df['Year'] = df['Order Date'].dt.year
df['Month'] = df['Order Date'].dt.month

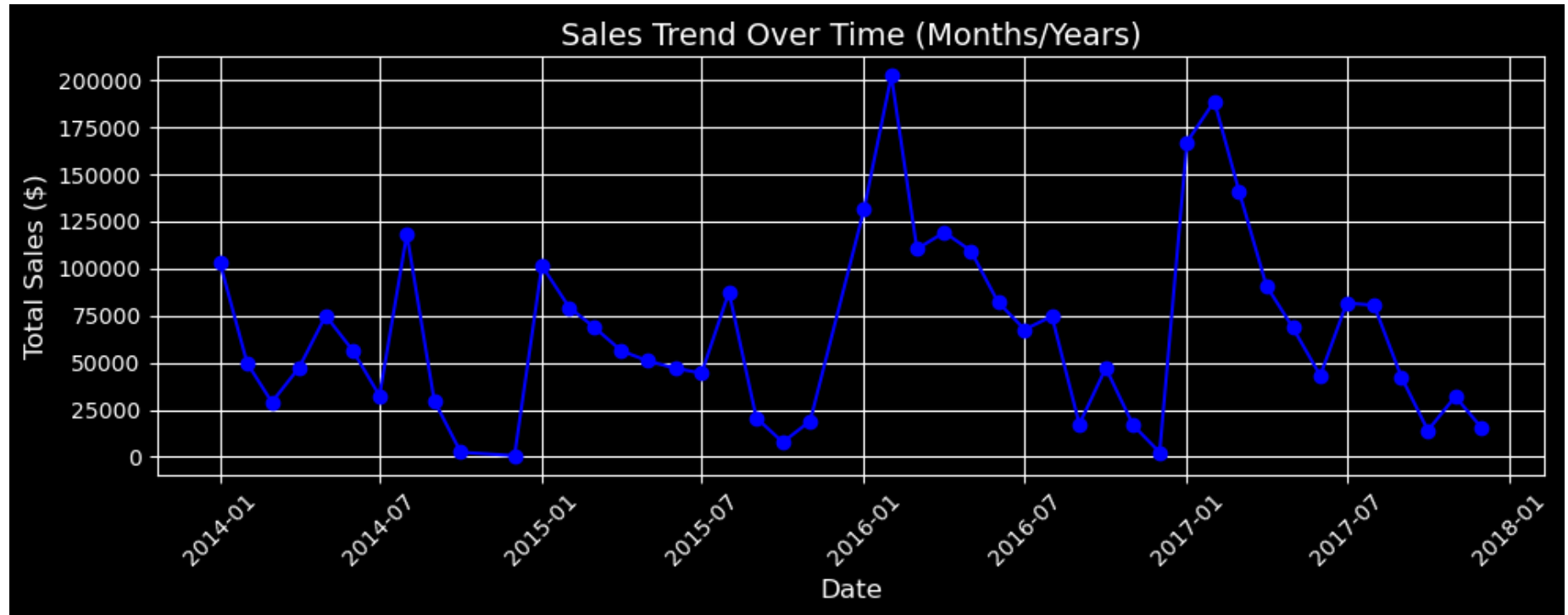
# Group by Year and Month to calculate total sales for each period
sales_trend = df.groupby(['Year', 'Month'])['Total Sales'].sum().reset_index()

# Create a 'Date' column for easier plotting
sales_trend['Date'] = pd.to_datetime(sales_trend['Year'].astype(str) + '-' + sales_trend['Month'].astype(str) + '-01')

# Plotting the sales trend over months/years
plt.figure(figsize=(10, 4))
plt.plot(sales_trend['Date'], sales_trend['Total Sales'], marker='o', color='b', label='Total Sales')

# Adding titles and labels
plt.title('Sales Trend Over Time (Months/Years)', fontsize=14)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Total Sales ($)', fontsize=12)
plt.xticks(rotation=45) # Rotate x-axis labels for better readability
plt.grid(True)
plt.tight_layout()
```

```
# Show the plot
plt.show()
```



```
In [77]: # Make a copy of the filtered DataFrame
year = df[df['Order Date'].dt.year.isin([2014])].copy()
# Add the 'Month' column
year['Month'] = year['Order Date'].dt.month
# Group by month and calculate sum of 'Total Sales' and 'Profit'
SalesProfit_by_Month = year.groupby('Month')[['Total Sales', 'Profit']].sum().round()
print(SalesProfit_by_Month)
```

Month	Total Sales	Profit
1	103464.0	3366.0
2	49761.0	2499.0
3	29024.0	-490.0
4	47464.0	3631.0
5	74988.0	1673.0
6	56283.0	1989.0
7	32429.0	955.0
8	118316.0	-602.0
9	29809.0	1081.0
10	2427.0	49.0
12	618.0	-332.0

```
In [150... # Which months have the highest sales?
# Convert 'Order Date' to datetime format if it's not already
df['Order Date'] = pd.to_datetime(df['Order Date'])

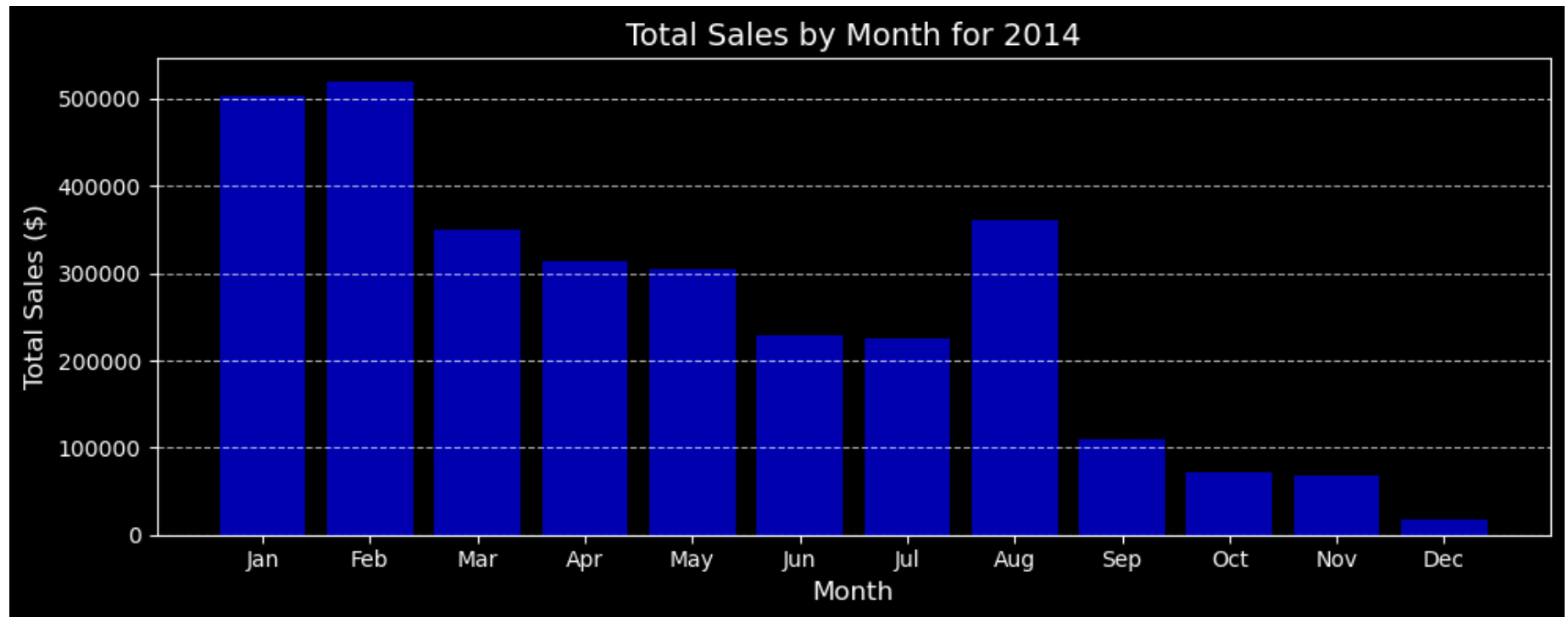
# Extract the 'Month' from 'Order Date'
df['Month'] = df['Order Date'].dt.month

# Group by Month to calculate total sales for each month
monthly_sales = df.groupby('Month')['Total Sales'].sum().reset_index()

# Plotting the total sales by month
plt.figure(figsize=(10, 4))
plt.bar(monthly_sales['Month'], monthly_sales['Total Sales'], color='b', alpha=0.7)

# Adding titles and labels
plt.title('Total Sales by Month for 2014', fontsize=14)
plt.xlabel('Month', fontsize=12)
plt.ylabel('Total Sales ($)', fontsize=12)
plt.xticks(monthly_sales['Month'], ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])
plt.grid(True, axis='y', linestyle='--', alpha=0.7)

# Show the plot
plt.tight_layout()
plt.show()
```



```
In [81]: # Make a copy of the filtered DataFrame
year = df[df['Order Date'].dt.year.isin([2015])].copy()
# Add the 'Month' column
year['Month'] = year['Order Date'].dt.month
# Group by month and calculate sum of 'Total Sales' and 'Profit'
SalesProfit_by_Month = year.groupby('Month')[['Total Sales', 'Profit']].sum().round()
print(SalesProfit_by_Month)
```


Month	Total Sales	Profit
1	101576.0	3004.0
2	79289.0	1.0
3	68945.0	672.0
4	56277.0	1543.0
5	50955.0	-158.0
6	47035.0	1810.0
7	44441.0	-947.0
8	87560.0	4888.0
9	20676.0	1117.0
10	7745.0	190.0
11	19017.0	435.0

In [148...

```
import plotly.express as px

# Make a copy of the filtered DataFrame
year = df[df['Order Date'].dt.year.isin([2015])].copy()

# Add the 'Month' column
year['Month'] = year['Order Date'].dt.month

# Group by month and calculate sum of 'Total Sales' and 'Profit'
SalesProfit_by_Month = year.groupby('Month')[['Total Sales', 'Profit']].sum().round()

# Create an interactive bar plot with Plotly
fig = px.bar(SalesProfit_by_Month,
              x=SalesProfit_by_Month.index, # 'Month' column as x-axis
              y=['Total Sales', 'Profit'], # Columns for y-axis
              labels={'Total Sales': 'Total Sales ($)', 'Profit': 'Profit ($)'},
              title="Total Sales and Profit by Month in 2015")

# Show the plot
fig.show()
```

```
In [84]: # Make a copy of the filtered DataFrame
year = df[df['Order Date'].dt.year.isin([2016])].copy()
# Add the 'Month' column
year['Month'] = year['Order Date'].dt.month
# Group by month and calculate sum of 'Total Sales' and 'Profit'
SalesProfit_by_Month = year.groupby('Month')[['Total Sales', 'Profit']].sum().round()
print(SalesProfit_by_Month)
```

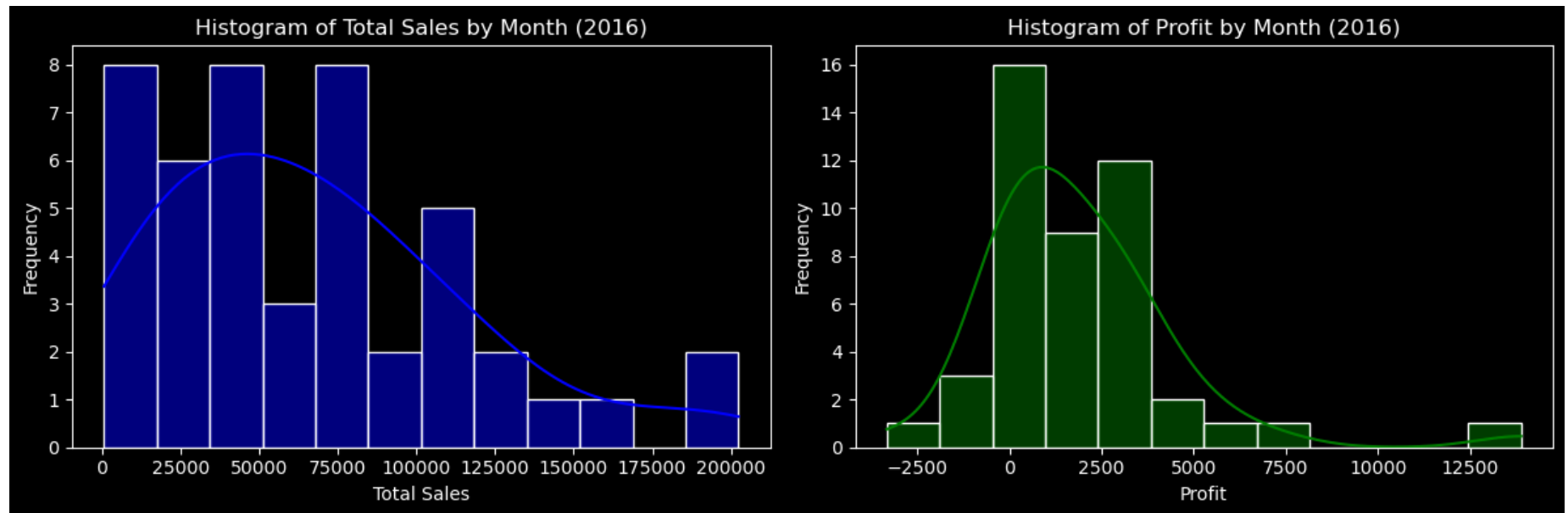
	Total Sales	Profit
Month		
1	131379.0	7057.0
2	202388.0	13891.0
3	110356.0	1519.0
4	119224.0	5032.0
5	109464.0	3226.0
6	82092.0	1384.0
7	67622.0	3616.0
8	74627.0	-248.0
9	17303.0	428.0
10	47204.0	505.0
11	16769.0	28.0
12	2365.0	66.0

```
In [144... # Plotting histograms for 'Total Sales' and 'Profit'
plt.figure(figsize=(12, 4))

# Histogram for 'Total Sales'
plt.subplot(1, 2, 1)
sns.histplot(SalesProfit_by_Month['Total Sales'], bins=12, kde=True, color='blue')
plt.title('Histogram of Total Sales by Month (2016)')
plt.xlabel('Total Sales')
plt.ylabel('Frequency')

# Histogram for 'Profit'
plt.subplot(1, 2, 2)
sns.histplot(SalesProfit_by_Month['Profit'], bins=12, kde=True, color='green')
plt.title('Histogram of Profit by Month (2016)')
plt.xlabel('Profit')
plt.ylabel('Frequency')

# Display the plots
plt.tight_layout()
plt.show()
```



```
In [86]: # Make a copy of the filtered DataFrame
year = df[df['Order Date'].dt.year.isin([2017])].copy()
# Add the 'Month' column
year['Month'] = year['Order Date'].dt.month
# Group by month and calculate sum of 'Total Sales' and 'Profit'
SalesProfit_by_Month = year.groupby('Month')[['Total Sales', 'Profit']].sum().round()
print(SalesProfit_by_Month)
```

	Total Sales	Profit
Month		
1	167176.0	3579.0
2	188890.0	3050.0
3	141064.0	5389.0
4	90845.0	-3332.0
5	68558.0	2757.0
6	43143.0	1562.0
7	81690.0	3093.0
8	80509.0	2939.0
9	42306.0	2483.0
10	14014.0	100.0
11	32024.0	953.0
12	15294.0	348.0

```
In [178... # Plotting line plot for 'Total Sales' and 'Profit'
plt.figure(figsize=(12, 4))

# Line plot for 'Total Sales'
sns.lineplot(data=SalesProfit_by_Month, x=SalesProfit_by_Month.index, y='Total Sales', marker='o', color='blue')

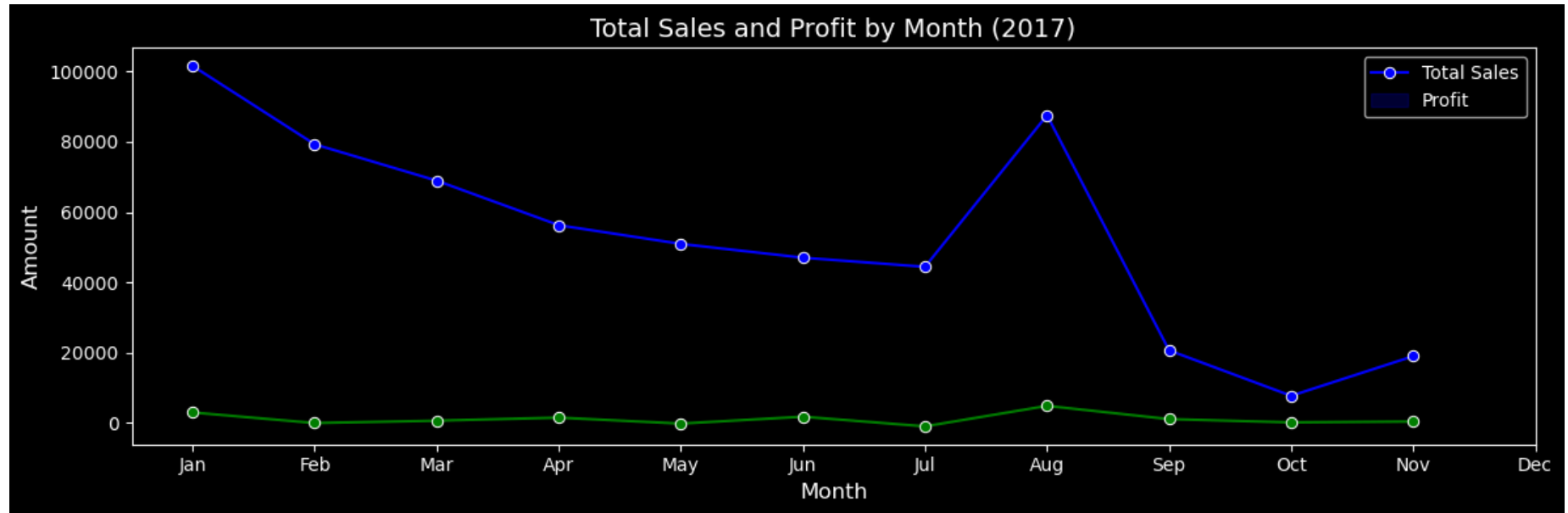
# Line plot for 'Profit'
sns.lineplot(data=SalesProfit_by_Month, x=SalesProfit_by_Month.index, y='Profit', marker='o', color='green')

# Adding labels and title
plt.title('Total Sales and Profit by Month (2017)', fontsize=14)
plt.xlabel('Month', fontsize=12)
plt.ylabel('Amount', fontsize=12)

# Setting x-axis labels for months
plt.xticks(ticks=range(1, 13), labels=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])

# Display the legend manually by specifying labels
plt.legend(labels=['Total Sales', 'Profit'])

# Show the plot
plt.tight_layout()
plt.show()
```



```
In [134... # Filter the data for specific years (2014, 2015, 2016, 2017)
year = df[df['Order Date'].dt.year.isin([2014, 2015, 2016, 2017])]

# Extract Month and Year from 'Order Date'
year['Month'] = year['Order Date'].dt.month
year['Year'] = year['Order Date'].dt.year

# Group by Year and Month and calculate total sales and profit
SalesProfit_by_Month = year.groupby(['Year', 'Month'])[['Total Sales', 'Profit']].sum().round()

# Print the result
print(SalesProfit_by_Month)
```

		Total Sales	Profit
Year	Month		
2014	1	103464.0	3366.0
	2	49761.0	2499.0
	3	29024.0	-490.0
	4	47464.0	3631.0
	5	74988.0	1673.0
	6	56283.0	1989.0
	7	32429.0	955.0
	8	118316.0	-602.0
	9	29809.0	1081.0
	10	2427.0	49.0
	12	618.0	-332.0
2015	1	101576.0	3004.0
	2	79289.0	1.0
	3	68945.0	672.0
	4	56277.0	1543.0
	5	50955.0	-158.0
	6	47035.0	1810.0
	7	44441.0	-947.0
	8	87560.0	4888.0
	9	20676.0	1117.0
	10	7745.0	190.0
	11	19017.0	435.0
2016	1	131379.0	7057.0
	2	202388.0	13891.0
	3	110356.0	1519.0
	4	119224.0	5032.0
	5	109464.0	3226.0
	6	82092.0	1384.0
	7	67622.0	3616.0
	8	74627.0	-248.0
	9	17303.0	428.0
	10	47204.0	505.0
	11	16769.0	28.0
	12	2365.0	66.0
2017	1	167176.0	3579.0
	2	188890.0	3050.0
	3	141064.0	5389.0
	4	90845.0	-3332.0
	5	68558.0	2757.0

6	43143.0	1562.0
7	81690.0	3093.0
8	80509.0	2939.0
9	42306.0	2483.0
10	14014.0	100.0
11	32024.0	953.0
12	15294.0	348.0

In [219...

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Filter the data for specific years (2014, 2015, 2016, 2017)
year = df[df['Order Date'].dt.year.isin([2014, 2015, 2016, 2017])]

# Extract Month and Year from 'Order Date'
year['Month'] = year['Order Date'].dt.month
year['Year'] = year['Order Date'].dt.year

# Group by Year and Month and calculate total sales and profit
SalesProfit_by_Month = year.groupby(['Year', 'Month'])[['Total Sales', 'Profit']].sum().round()

# Reset index to avoid conflicts with index columns
SalesProfit_by_Month = SalesProfit_by_Month.reset_index()

# Set dark background style
plt.style.use('dark_background')

# Create a scatter plot for 'Total Sales' vs 'Profit' with Year as hue
plt.figure(figsize=(10, 6))

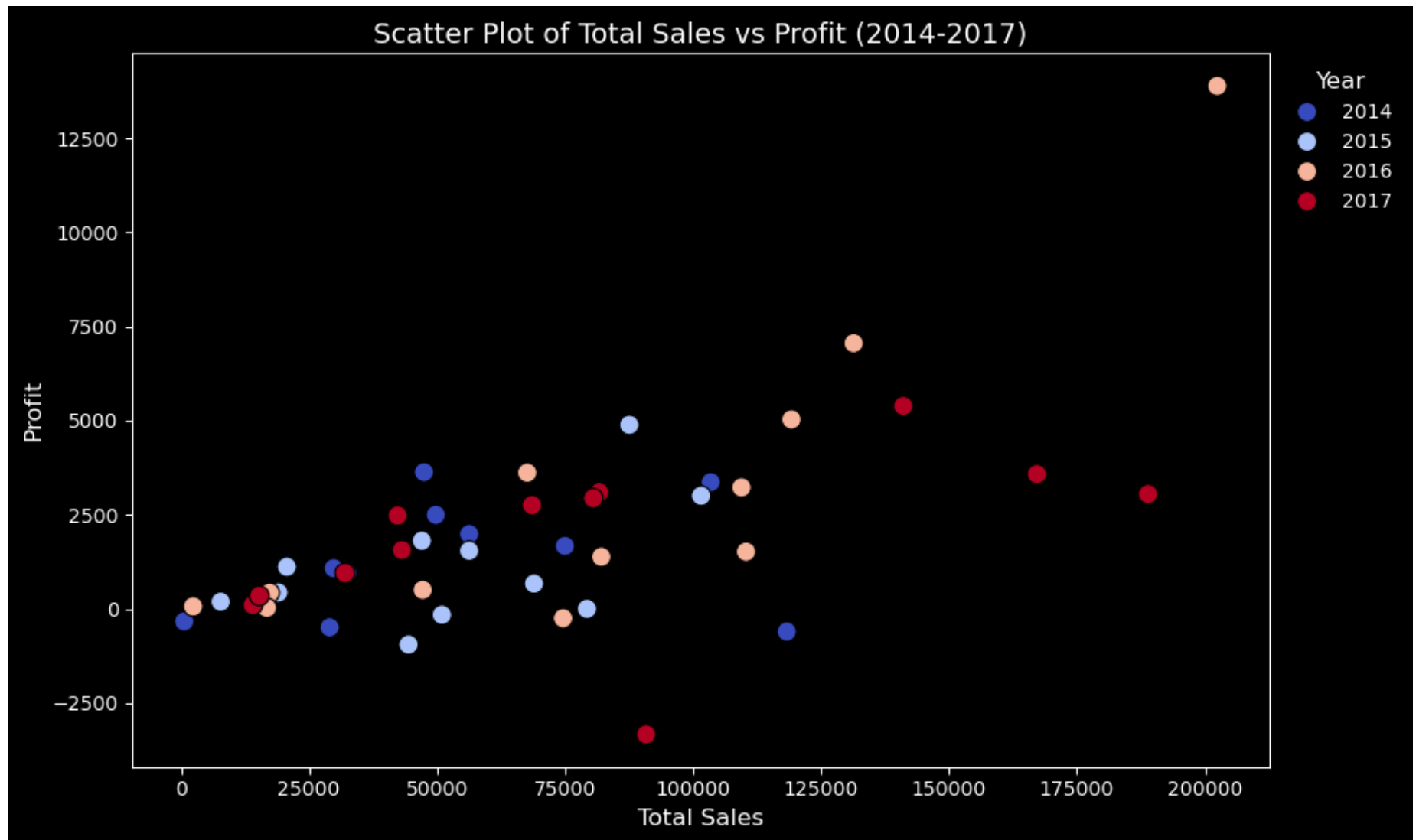
# Scatter plot using Seaborn with 'Year' as the hue
sns.scatterplot(data=SalesProfit_by_Month, x='Total Sales', y='Profit', hue='Year', palette='coolwarm', s=100, edgecolor='black')

# Adding Labels and title
plt.title('Scatter Plot of Total Sales vs Profit (2014-2017)', fontsize=14, color='white')
plt.xlabel('Total Sales', fontsize=12, color='white')
plt.ylabel('Profit', fontsize=12, color='white')

# Display the plot
plt.legend(title='Year', loc='upper left', bbox_to_anchor=(1, 1), title_fontsize='12', fontsize='10', frameon=False)
```



```
plt.tight_layout()  
plt.show()
```



Customer Analysis

```
In [276... # What is the total sales and profit by customer (Customer ID or Customer Name)?
# Group by Customer ID to calculate total sales and profit
customer_sales_profit = df.groupby(['Customer ID', 'Customer Name'])[['Total Sales', 'Profit']].sum().reset_index()

# Sorting by Total Sales and Profit to see top customers
customer_sales_profit_sorted_sales = customer_sales_profit.sort_values(by='Total Sales', ascending=False)
customer_sales_profit_sorted_profit = customer_sales_profit.sort_values(by='Profit', ascending=False)
print(customer_sales_profit_sorted_sales)
print(customer_sales_profit_sorted_profit)
```

	Customer ID	Customer Name	Total Sales	Profit
623	TC-20980	Tamara Chand	89645.374	8745.0635
75	BM-11140	Becky Martin	85675.036	-1759.8593
419	ME-17320	Maria Etezadi	57569.132	1754.3099
197	DR-12940	Daniel Raglin	56401.910	3050.3344
359	KF-16285	Karen Ferguson	51899.202	748.8844
..	...	...	...	...
50	AW-10930	Arthur Wiediger	9.656	1.7248
30	AI-10855	Arianne Irving	6.680	3.2064
526	RB-19705	Roger Barcio	4.982	-3.7880
472	NF-18475	Neil Französisch	4.710	0.0000
641	TS-21340	Toby Swindell	1.424	-1.8904

[669 rows x 4 columns]

	Customer ID	Customer Name	Total Sales	Profit
623	TC-20980	Tamara Chand	89645.374	8745.0635
197	DR-12940	Daniel Raglin	56401.910	3050.3344
574	SE-20110	Sanjit Engle	44171.780	2825.2862
356	KD-16495	Keith Dawkins	22288.578	2632.0931
478	NM-18445	Nathan Mautz	12987.304	2390.8490
..	...	...	...	...
136	CP-12340	Christine Phan	7013.315	-1046.2325
495	PF-19120	Peter Fuller	6099.196	-1115.6038
75	BM-11140	Becky Martin	85675.036	-1759.8593
667	ZC-21910	Zuschuss Carroll	8675.806	-1778.0546
264	GT-14635	Grant Thornton	36840.832	-4108.6589

[669 rows x 4 columns]

```
In [269... # Which customers have made the highest number of orders?
# Group by Customer ID and count the number of unique Order IDs for each customer
customer_order_count = df.groupby(['Customer Name', 'Customer ID'])['Order ID'].nunique().reset_index()

# Rename the column to "Number of Orders"
customer_order_count = customer_order_count.rename(columns={'Order ID': 'Number of Orders'})

# Sort by Number of Orders in descending order
Top5Customerby_OrderCount = customer_order_count.sort_values(by='Number of Orders', ascending=False)
print(Top5Customerby_OrderCount.head(10))
```

	Customer Name	Customer ID	Number of Orders
581	Shahid Hopkins	SH-20395	7
484	Noel Staavos	NS-18640	7
561	Sam Zeldin	SZ-20035	6
120	Chris Cortes	CC-12220	6
545	Roland Schwarz	RS-19765	5
257	George Bell	GB-14530	5
412	Maribeth Schnelling	MS-17365	5
356	Katherine Nockton	KN-16390	5
536	Rob Dowd	RD-19585	5
425	Matt Collins	MC-17575	5

```
In [95]: # Group by City and State to calculate total sales
city_state_sales = df.groupby(['City', 'State'])['Total Sales'].sum().reset_index()

# Sort by Total Sales in descending order to identify the highest sales locations
city_state_sales_sorted = city_state_sales.sort_values(by='Total Sales', ascending=False).round()
print(city_state_sales_sorted.head(10))
```

	City	State	Total Sales
234	New York City	New York	331850.0
192	Los Angeles	California	253904.0
309	San Francisco	California	149115.0
317	Seattle	Washington	114783.0
307	San Antonio	Texas	98314.0
263	Philadelphia	Pennsylvania	94712.0
148	Houston	Texas	91197.0
168	Lafayette	Indiana	89345.0
140	Henderson	Kentucky	65246.0
272	Providence	Rhode Island	64073.0

```
In [138... # DataFrame = 'df' and it contains the relevant columns ('City', 'State', 'Total Sales')
# Group by City and State to calculate total sales
city_state_sales = df.groupby(['City', 'State'])['Total Sales'].sum().reset_index()
# Sort by Total Sales in descending order
city_state_sales_sorted = city_state_sales.sort_values(by='Total Sales', ascending=False).round()

# Create a treemap
fig = px.treemap(city_state_sales_sorted,
                  path=['State', 'City'], # Hierarchy: State -> City
                  values='Total Sales', # Values used for the size of the rectangles
                  color='Total Sales', # Color based on Total Sales
                  hover_data=['Total Sales'], # Display Total Sales on hover
                  title="Total Sales by City and State")

fig.show()
```

```
In [98]: # How many products in each sub-category are sold in total (Quantity)?  
# Group by Sub-Category and calculate the total Quantity sold  
sub_category_quantity = df.groupby('Sub-Category')['Quantity'].sum().reset_index()  
  
# Sort by Quantity sold in descending order  
sub_category_quantity_sorted = sub_category_quantity.sort_values(by='Quantity', ascending=False)
```

```
# Print the sub-categories and their total quantities sold
print(sub_category_quantity_sorted)
```

	Sub-Category	Quantity
3	Binders	1716
12	Paper	1392
9	Furnishings	975
14	Storage	863
0	Accessories	834
13	Phones	817
2	Art	801
5	Chairs	634
1	Appliances	468
10	Labels	404
16	Tables	338
8	Fasteners	233
7	Envelopes	225
4	Bookcases	225
15	Supplies	167
11	Machines	152
6	Copiers	76

```
In [99]: # 'Quantity' data across different 'Sub-Categories' and 'Regions'
heatmap_data = df.pivot_table(values='Quantity', index='Sub-Category', columns='Region', aggfunc='sum')

# Create the heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(heatmap_data, annot=True, cmap='Blues', fmt='d')
plt.title('Total Quantity Sold by Sub-Category and Region', fontsize=16)
plt.show()
```



```
In [193... df.groupby(['Category', 'Sub-Category'])['Order ID'].nunique().reset_index()
```

Out[193...

	Category	Sub-Category	Order ID
0	Furniture	Bookcases	56
1	Furniture	Chairs	159
2	Furniture	Furnishings	238
3	Furniture	Tables	89
4	Office Supplies	Appliances	120
5	Office Supplies	Art	196
6	Office Supplies	Binders	356
7	Office Supplies	Envelopes	64
8	Office Supplies	Fasteners	57
9	Office Supplies	Labels	104
10	Office Supplies	Paper	325
11	Office Supplies	Storage	212
12	Office Supplies	Supplies	43
13	Technology	Accessories	201
14	Technology	Copiers	24
15	Technology	Machines	38
16	Technology	Phones	214

Visualizations to Support Insights:

Trend Analysis: Line chart, Histogram for sales and profit over time (yearly and monthly trends). Heat map for Total Quantity Sold by Sub-Category and Region Tree Map used for Total Sales by City and State. Scatter Plot of Total Sales vs Profit from 2014-2017. Stem Plot for total

sales & profit by each region Area chart used for sum of 'Total Sales' and 'Profit' by city. Sunburst plot or donut chart used for customer segment (Consumer, Corporate, Home Office). Bar chart for Average Total Sales & Profit. Pie chart used for Total Sales and Profit by Ship mode & product category generates the most sales and profit.

Conclusion:

Sales and Profit Performance: Total sales reached \$3,074,404 & profit \$85801.0 but the profit margin is relatively low at 2.8%. Technology is the most profitable category, despite Furniture driving the highest sales. Sub-category insights show Copiers as the top-performing sub-category in terms of both sales and profit, while Fasteners significantly underperform.

Customer Insights: Consumers contribute the largest share of sales and profit, followed by the Corporate and Home Office segments. Key customers like Tamara Chand drive substantial sales and profit, emphasizing the importance of customer retention. Negative profitability for certain customers highlights areas where pricing or discount strategies may need adjustment.

Regional Insights: California leads in sales and profit, while New York City stands out as the most profitable city. The West region dominates in total sales, while the East region achieves the highest profitability. There are opportunities to grow sales in underperforming regions.

Product and Shipping Mode Analysis: High-performing products, such as the Canon imageCLASS Copier, drive significant sales and profit, whereas low-performing products and sub-categories need optimization or repositioning. The Standard Class shipping mode generates the highest sales and profit, but diversifying and promoting alternative shipping methods could increase customer satisfaction.

Temporal Trends: Sales peaked in 2016, followed by a slight decline in 2017, which requires further investigation. Monthly trends indicate January 2015 as a high-sales period, suggesting potential for seasonal campaigns.

In []: